Recall:

A Subnormal series of G:

$$\{e\} = H_0 \triangleleft H_1 \triangleleft H_2 \triangleleft \cdots \triangleleft H_{n-1} \triangleleft H_n = G$$

A Normal series of G: $H_i \trianglelefteq G$,

$$\{e\} = H_0 < H_1 < H_2 < \cdots < H_{n-1} < H_n = G$$

Normal $\underset{\not\Leftarrow}{\Rightarrow}$ Subnormal.

「Butterfly Lemma.

$H, K < G$, $H^* \triangleleft H$, $K^* \triangleleft K$.

1). $H \cap K$, $H^* \cap K$, $H \cap K^* < H, K$

2). $H^*(H \cap K)$, $H^*(H \cap K^*)$, $K^*(H \cap K)$, $K^*(H^* \cap K) \leq G$

3). $H^* \cap K$, $H \cap K^* \triangleleft H \cap K$.

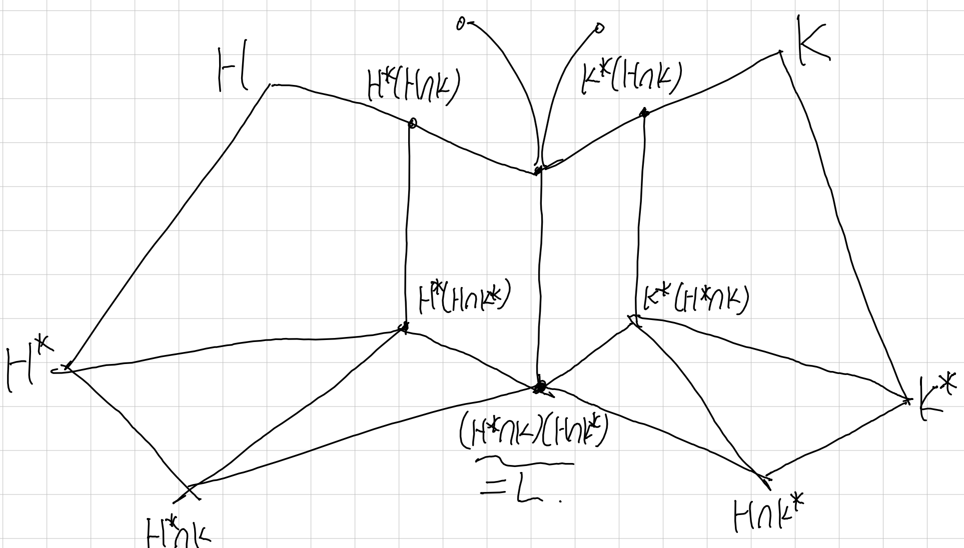
$H^*(H \cap K^*) \triangleleft H^*(H \cap K)$

$K^*(H^* \cap K) \triangleleft K^*(H \cap K)$

$H^*(H \cap K) / H^*(H \cap K^*)$

$\simeq K^*(H \cap K) / K^*(H^* \cap K)$

$\simeq H \cap K / L$



#4.

Consider refinements of given series respectively:

$1 < \langle 18 \rangle < \langle 6 \rangle < \langle 3 \rangle < \mathbb{Z}_{72}$
$\quad\ 4 \qquad 12 \qquad 24$

and

$1 < \langle 24 \rangle < \langle 12 \rangle < \langle 3 \rangle < \mathbb{Z}_{72}$

$\quad 3 \qquad 6 \qquad 24$

Then $\langle 18 \rangle / 1 \cong \langle 3 \rangle / \langle 12 \rangle$,

$\langle 6 \rangle / \langle 18 \rangle \cong \langle 24 \rangle / 1$

$\langle 3 \rangle / \langle 6 \rangle \cong \langle 12 \rangle / \langle 24 \rangle$.

#9.

S_3
|
$A_3 (= \mathbb{Z}_3)$
|
1

$1 \times 1 \triangleleft 1 \times \mathbb{Z}_2 \triangleleft A_3 \times \mathbb{Z}_2 \triangleleft S_3 \times \mathbb{Z}_2$

$1 \times 1 \triangleleft A_3 \times 1 \triangleleft S_3 \times 1 \triangleleft S_3 \times \mathbb{Z}_2$

$1 \times 1 \triangleleft A_3 \times 1 \triangleleft A_3 \times \mathbb{Z}_2 \triangleleft S_3 \times \mathbb{Z}_2$

All factor group in above three series has order 3 or 2, thus simple.

#12. Since the operation is defined pointwise,

$Z(S_3 \times D_4) = Z(S_3) \times Z(D_4)$

$= \{1\} \times \{1, \rho_2\}$.

$\varphi : S_3 \times D_4 \to S_3 \times K_4$

$: (a, b) \to$

#14.

$S_3 \times D_4 / \{1\} \times \{1, \rho^2\} \cong (S_3/\{1\}) \times (D_4 / \{1, \rho_2\})$

$\cong S_3 \times K_4$ (K_4 is Klein-four group)

Since K_4 is Abelian, $Z(S_3 \times K_4) = \{1\} \times K_4$. Now, the ascending central series is

$1 \times 1 < \{1\} \times \{1, \rho_2\} < \{1\} \times D_4 \leq \{1\} \times D_4 \leq \cdots$

#18.

$$1 \triangleleft A_3 \times 1 \triangleleft A_3 \times A_3 \triangleleft S_3 \times A_3 \triangleleft S_3 \times S_3$$

is Composition series, because all factor group has order of 2 or 3, thus Simple.

(Normality is shown by $[S_3 : A_3] = 2$, thus Normal)

Moreover, group of order 2 or 3 is only $\mathbb{Z}_2$ or $\mathbb{Z}_3$, respectively, it is abelian.

Therefore $S_3 \times S_3$ is solvable.

#19.

$1 \triangleleft \mathbb{Z}_2 \triangleleft \mathbb{Z}_4 \triangleleft D_4$ is Composition series of D_4, $\left(\begin{array}{l} \mathbb{Z}_2 = \{1, \rho_2\}, \\ \mathbb{Z}_4 = \{1, \rho_2, \rho_3, \rho_4\} \end{array} \right)$

because $|D_4 : \mathbb{Z}_4| = |\mathbb{Z}_4 : \mathbb{Z}_2| = |\mathbb{Z}_2 : 1| = 2$,

thus Normal, and all factor group is simple, abelian

#20. Since cyclic group is Abelian, every subgroup is Normal.

By Schreier Thm, (1) and (2) have isomorphic refinement as:

$$\underset{H_0}{1} < \underset{H_1}{\langle 12 \rangle} < \underset{H_2}{\langle 3 \rangle} < \underset{H_3}{\mathbb{Z}_{36}}, \qquad \underset{K_0}{1} < \underset{K_1}{\langle 18 \rangle} < \underset{K_2}{\mathbb{Z}_{36}}.$$

$$1 = H_0 \le \underset{1}{\underset{\parallel}{1 \cdot (\langle 12 \rangle \cap 1)}} < \underset{1}{\underset{\parallel}{1 \cdot (\langle 12 \rangle \cap \langle 18 \rangle)}} < 1 \cdot (\langle 12 \rangle \cap \mathbb{Z}_{36}) = \langle 12 \rangle$$

$$= H_1 \le \underset{\langle 12 \rangle}{\underset{\parallel}{H_1 \cdot (\langle 3 \rangle \cap 1)}} < \underset{\langle 12 \rangle (\langle 18 \rangle) = \langle 6 \rangle}{\underset{\parallel}{H_1 (\langle 3 \rangle \cap \langle 18 \rangle)}} \le H_1 (\langle 3 \rangle \cap \mathbb{Z}_{36}) = \langle 3 \rangle$$

$$= H_2 \le \underset{\langle 3 \rangle}{\underset{\parallel}{H_2 \cdot 1}} \le \underset{\langle 3 \rangle}{\underset{\parallel}{H_2 \cdot \langle 18 \rangle}} \le H_2 \cdot \mathbb{Z}_{36} = \mathbb{Z}_{36} = H_3.$$

Similarly,

$$K_0 \le \underset{1}{\underset{\parallel}{K_0 (\langle 18 \rangle \cap 1)}} \le \underset{1}{\underset{\parallel}{K_0 (\langle 18 \rangle \cap \langle 12 \rangle)}} \le \underset{\langle 18 \rangle}{\underset{\parallel}{K_0 (\langle 18 \rangle \cap \langle 3 \rangle)}}$$

$$\le K_1 \le \underset{\langle 18 \rangle}{\underset{\parallel}{K_1 (\mathbb{Z}_{36} \cap 1)}} \le \underset{\langle 6 \rangle}{\underset{\parallel}{K_1 (\mathbb{Z}_{36} \cap \langle 12 \rangle)}} \le \underset{\langle 3 \rangle}{\underset{\parallel}{K_1 (\mathbb{Z}_{36} \cap \langle 3 \rangle)}}$$

$$\le \mathbb{Z}_{36}.$$

Now, we obtained two refinement $\left\{ \begin{array}{l} 1 \triangleleft \overset{①}{\langle 12 \rangle} \triangleleft \overset{②}{\langle 6 \rangle} \triangleleft \overset{③}{\langle 3 \rangle} \triangleleft \overset{④}{\mathbb{Z}_{36}} \\ 1 \triangleleft \overset{⑤}{\langle 18 \rangle} \triangleleft \overset{⑥}{\langle 6 \rangle} \triangleleft \overset{⑦}{\langle 3 \rangle} \triangleleft \overset{⑧}{\mathbb{Z}_{36}}. \end{array} \right.$

with Correspondences ④ $\simeq$ ⑧, ② $\simeq$ ⑤, ① $\simeq$ ⑥, ③ $\simeq$ ⑦.

#23. By Lagrange Thm,

$$|G| = |H_n : H_{n-1}| \cdot |H_{n-1}|$$
$$= |H_n : H_{n-1}| \cdot |H_{n-1} : H_{n-2}| \cdot |H_{n-2}|$$
$$= \quad \vdots$$
$$= \prod_{i=0}^{n-1} |H_{i+1} : H_i| = \prod_{i=0}^{n-1} |H_{i+1}/H_i| = s_1 \cdots s_n.$$

#24.

Consider the Rational Numbers group $(\mathbb{Q}, +)$.

Since G/N is simple iff N is Maximal Normal,

If $\mathbb{Q}$ has a Composition Series, then $\mathbb{Q}$ has a Maximal Normal Subgroup.

But, suppose $N \triangleleft \mathbb{Q}$ is Maximal, then

$$N \subsetneq \frac{1}{2}N \subsetneq \mathbb{Q},$$

This is contradiction. That is, $\mathbb{Q}$ has no Maximal Normal Subgroup.

As a result, $\mathbb{Q}$ has no Composition Series.

#28. For each i, define a map:

$$\varphi : H_i N \to \gamma[H_i]/\gamma[H_{i-1}] : hn \mapsto hnN\gamma[H_{i-1}]$$

Well-defined: for $h_1, h_2 \in H_i$, $n_1, n_2 \in N$, suppose $h_1 n_1 = h_2 n_2$. Then,

$$h_1 n_1 N\gamma[H_{i-1}] \underset{n_1 \in N}{=} h_1 N\gamma[H_{i-1}] \underset{n_2 \in N}{=} h_2 n_2 n_1^{-1} N\gamma[H_{i-1}] = h_2 n_2 N\gamma[H_{i-1}]$$

Thus, well-defined.

Homomorphism: for any $h_1, h_2 \in H_i$, $n_1, n_2 \in N$,

$$\begin{aligned}
\varphi(h_1 n_1 \cdot h_2 n_2) &= \varphi(h_1 h_2 n_3 n_2) \quad \text{for some } n_3 \in N \ (\text{because } N \lhd G) \\
&= h_1 h_2 n_3 n_2 N\gamma[H_{i-1}] \\
&= h_1 h_2 N\gamma[H_{i-1}] \\
&= (h_1 N\gamma[H_{i-1}])(h_2 N\gamma[H_{i-1}]) = \varphi(h_1 n_1)\cdot\varphi(h_2 n_2)
\end{aligned}$$

Moreover, $\ker\varphi = \{hn \in H_i N \mid hnN\gamma[H_{i-1}] = 1\cdot N\gamma[H_{i-1}]\}$

$= H_{i-1}N$ because $hnN\gamma[H_{i-1}] = 1\cdot N\cdot\gamma[H_{i-1}]$

$\Leftrightarrow h \in H_{i-1}$ and $n \in N$.)

Onto is trivial.

Now, the first isomorphism Thm gives

$$(H_i N)/(H_{i-1}N) \overset{1st}{\cong} \gamma[H_i]/\gamma[H_{i-1}]$$

Excercise 27 gives us that $H_i N/H_i N$ simple,

so is $\gamma[H_i]/\gamma[H_{i-1}]$. Therefore $\{\gamma[H_i]\}$ is a composition series for G.